

LETHBRIDGE DEMO FARM AGDM, AB, Canada

WMO: 715090

Lat: 49.6867N

Lon: 112.7449W

Elev: 907

StdP: 90.89

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-27.5	-24.2	-30.7	0.2	-27.4	-27.1	0.3	-24.1	16.7	4.7	15.5	4.6	2.7	0	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	30.1	16.6	28.4	16.2	26.8	15.7	19.2	25.9	18.0	25.2	16.9	24.4	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	13.3	23.2	15.3	12.1	21.6	14.0	11.1	20.2	58.8	25.9	54.5	25.1	50.8	24.6	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.5	11.1	-33.5	33.7	3.0	1.9	-35.7	35.0	-37.4	36.1	-39.1	37.1	-41.3	38.5		
(5)			-33.3	22.0	3.2	1.6	-35.6	23.1	-37.4	24.0	-39.2	24.9	-41.6	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-4.2	-5.3	0.4	5.6	11.2	15.0	18.2	17.1	12.8	6.5	0.2	-5.7
	DBStd	10.40	8.56	8.69	7.19	4.96	4.28	3.07	2.67	3.10	4.46	5.25	7.34	8.38
	HDD10.0	2288	439	430	300	144	37	2	0	1	23	131	296	486
	HDD18.3	4551	697	663	557	382	224	108	35	59	169	368	544	744
	CDD10.0	847	1	1	2	12	74	151	256	222	108	21	2	0
	CDD18.3	68	0	0	0	0	2	7	32	22	5	0	0	0
	CDH23.3	1417	0	0	0	9	90	141	544	448	180	4	0	0
	CDH26.7	377	0	0	0	1	16	26	151	136	47	0	0	0
Wind	WSAvg	4.6	5.7	4.6	4.9	5.1	4.5	4.4	3.6	3.4	3.9	4.8	5.5	5.3
Precipitation	PrecAvg	425	24	16	28	40	54	67	45	44	46	18	18	23
	PrecMax	766	59	37	78	169	134	169	123	143	140	46	51	58
	PrecMin	243	5	2	5	0	10	3	7	4	4	0	0	4
	PrecStd	111	13	8	16	33	30	42	32	32	35	13	14	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	13.8	17.7	24.7	28.8	30.0	32.4	32.6	30.6	23.5	18.1	12.0
		MCWB	6.6	5.4	8.0	11.2	13.9	17.3	17.6	16.3	15.0	11.8	9.0	5.3
	2%	DB	10.5	10.7	15.0	20.4	26.0	26.9	30.0	29.9	27.9	20.2	14.1	9.4
		MCWB	5.0	4.3	7.0	9.4	13.3	15.7	17.9	16.1	14.2	10.8	6.9	4.4
	5%	DB	8.2	8.3	12.8	17.4	23.2	24.8	28.3	28.0	25.3	17.4	11.7	7.4
		MCWB	3.5	2.9	5.7	8.0	12.0	15.4	17.6	15.7	13.4	9.5	5.9	3.1
	10%	DB	6.4	6.1	10.4	14.7	20.6	22.8	26.5	25.9	22.2	14.9	9.6	5.6
		MCWB	2.4	1.6	4.5	6.7	10.9	14.0	16.9	15.3	12.5	8.1	4.5	1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	6.6	8.9	11.7	14.8	19.7	21.8	19.4	16.6	13.0	9.7	6.5
		MCDB	12.4	12.0	16.0	22.8	25.3	26.3	28.7	26.4	26.0	21.0	17.1	11.1
	2%	WB	5.1	4.6	7.4	9.9	13.6	17.5	20.1	18.0	15.0	11.4	7.7	4.7
		MCDB	10.1	10.1	14.0	19.6	24.1	24.5	27.0	26.3	25.1	18.7	13.5	8.8
	5%	WB	3.7	3.1	6.0	8.4	12.5	16.0	18.7	16.9	14.0	9.9	6.1	3.3
		MCDB	7.9	8.0	12.1	16.4	21.8	22.7	25.3	25.1	24.0	16.6	11.1	7.0
	10%	WB	2.6	1.8	4.7	7.0	11.5	14.8	17.7	16.0	12.8	8.4	4.8	1.9
		MCDB	6.2	5.9	10.0	14.0	19.4	21.6	24.6	23.9	21.6	14.3	9.4	5.7
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.2	11.6	12.6	13.1	12.1	14.3	14.6	14.0	12.2	10.9	10.2
		MCDBR	10.8	13.3	15.4	18.2	18.4	16.4	18.1	19.6	20.3	16.1	13.4	10.6
		MCWBR	7.0	8.2	8.8	9.1	7.9	7.6	8.5	8.2	8.5	8.3	7.9	7.0
		MCDDBR	10.8	13.1	14.9	17.0	16.5	14.7	15.7	16.6	18.7	15.5	13.0	11.0
	5% WB	MCWBR	7.3	8.5	8.9	8.9	7.5	7.5	8.6	7.9	8.2	8.3	8.0	7.7
Clear-Sky Solar Irradiance	taub	0.251	0.259	0.284	0.320	0.333	0.330	0.333	0.350	0.312	0.280	0.256	0.234	
	taud	2.459	2.491	2.483	2.391	2.406	2.460	2.467	2.395	2.473	2.541	2.505	2.458	
	Ebn at Noon	831	912	938	928	925	927	919	885	891	871	817	805	
	Edh at Noon	58	73	88	108	111	106	104	107	88	69	55	50	
All-Sky Solar Radiation	RadAvg	1.24	2.24	3.50	4.84	5.77	6.42	7.05	5.61	3.98	2.48	1.38	0.98	
	RadStd	0.10	0.17	0.29	0.33	0.48	0.46	0.36	0.34	0.33	0.25	0.11	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page